ARTIFICIAL INTELLIGENCE · DIGITAL ASSIGNMENT (CASE STUDY)

Interactive 8-Puzzle Solver Depth Limited Search

STUDENT
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REGISTRATION NUMBER
24BCT0058

ASSIGNED ALGORITHM
03 · DLS

Project and submission links

GitHub repository
https://github.com/skapyskar/ai-vit-da

GitHub Pages live website
https://skapyskar.github.io/ai-vit-da/

README, screenshots and run instructions
https://github.com/skapyskar/ai-vit-da/blob/main/README.md

Explanatory demonstration video (synthetic narration)
https://skapyskar.github.io/ai-vit-da/docs/24BCT0058-dls-demo.mp4

The registration allocation specifies DLS only. This project implements DLS from scratch and compares independent depth bounds. DLS is uninformed and uses no heuristic.

Features demonstrated

Manual and random solvable puzzles; custom goals; animated moves; parent, successors, frontier and expanded history; pause/resume and stepping; solution replay; measured statistics; depth-limit comparisons; JSON export; and responsive layouts.

Prepared for Aayush Dwivedi, 24BCT0058. HTML5, CSS3, JavaScript and Bootstrap. OpenAI Codex assisted with implementation and documentation. The walkthrough uses synthetic speech; review the source and demonstration before submission.